

CMSC 10 – Introduction to Computer Science

History of Computer/ Computing

Week 2/16

History of Computing

Abacus (2700 – 2300 BC)

The abacus is believed to have been invented in ancient Mesopotamia between 2700 and 2300 BC.

It was a simple counting tool that used beads on rods to represent numbers.



History of Computing

500 BC

Panini: the Indian Mathematician introduces the forerunner to modern formal language theory

300 BC

Pingala: invents the binary number system

87 BC

Antikythera Mechanism: Built in Rhodes to track movement of the stars

60 AD

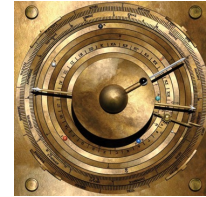
Heron of Alexandria: invents machines which are able to follow a series of instructions

724

Liang Ling-Can: invents the first fully mechanical clock

1492

Leonardo da Vinci: Drawings by Leonardo da Vinci depict inventions such as flying machines, including a helicopter, the first mechanical calculator and one of the first programmable robots



The Antikythera mechanism is an Ancient Greek hand powered orrery (model of the Solar System), described as the oldest known example of an analogue computer used to predict astronomical positions and eclipses decades in advance.

History of Computing

1614

John Napier: invents a system of moveable rods (Napier's Rods) based on logarithms which were able to multiply, divide and calculate square and cube roots

1622

William Oughtred: develops the slide rule (it was circular in shape)

1623

Calculating Clock: is invented by Wilhelm Schickard

1642

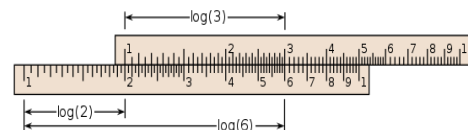
Blaise Pascal: invents the the "Pascaline", a mechanical adding machine

1671

Gottfried Leibniz: becomes known as one of the founding fathers of calculus



<https://collection.sciencemuseumgroup.org.uk/>



<https://en.wikipedia.org/>

History of Computing

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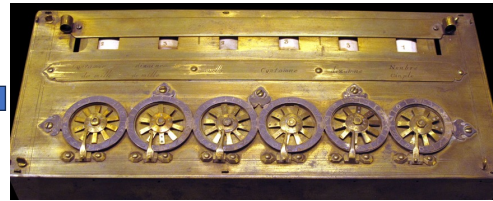
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Pascal Machine @ <https://en.wikipedia.org/>

History of Computing

1801

Joseph-Marie Jacquard: invents an automatic loom controlled by punched cards

1820

Arithmometer: The Arithmometer was the first mass-produced calculator invented by Charles Xavier Thomas de Colmar

1822

Charles Babbage: designs his first mechanical computer

1834

Analytical Engine: The Analytical Engine was invented by Charles Babbage

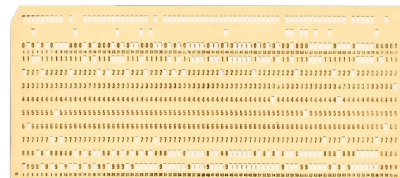
1835

Morse code: is invented by Samuel Morse

1848

Boolean algebra: is devised by George Boole

Example of a punch card



Source: computerhope.com/jargon/p/punccard.htm

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The arithmometer (French: arithmomètre) was the first digital mechanical calculator strong enough and reliable enough to be used daily in an office environment.

This calculator could add and subtract two numbers directly and could perform long multiplications and divisions effectively by using a movable accumulator for the result.

<https://en.wikipedia.org/>

History of Computing

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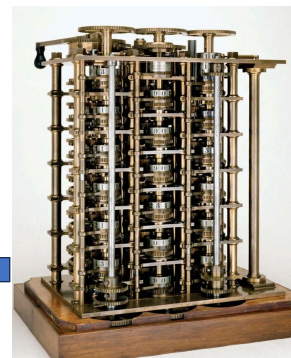
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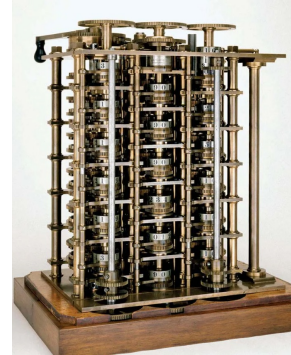


First Mechanical Computer/ Difference Engine
<https://www.britannica.com/technology/Difference-Engine#/media/1/725539/19223>

History of Computing

First Mechanical Computer/ Difference Engine

- Invented by Charles Babbage around 1820s to 1830s
 - An English mathematician and inventor
 - a pioneer in the fields of operations research and actuarial science.
- Difference Engine, an early calculating machine, verging on being the first computer, designed and partially built during the 1820s and '30s.



First Mechanical Computer/ Difference Engine
<https://www.britannica.com/technology/Difference-Engine#/media/1/725539/19223>

History of Computing

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<https://www.britannica.com/topic/Morse-Code#/media/1/393067/229802>



History of Computing

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The Codes

History of Computing

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Boolean algebra

- symbolic system of mathematical logic that represents relationships between entities—either ideas or objects.
- formulated in 1847 by George Boole of England and were subsequently refined by other mathematicians and applied to set theory
- It is of significance to the theory of probability, geometry of sets, and information theory.
- it constitutes the basis for the design of circuits used in electronic digital computers.

History of Computing

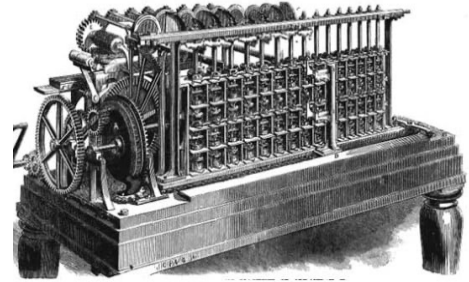
1853 **Tabulating Machine:** is invented by Per Georg Scheutz and his son Edvard

1869 **William Stanley Jevons:** designs a practical logic machine

1878 **Ramon Verea:** invents a fast calculator with an internal multiplication table

1880 **Alexander Graham Bell:** invents the telephone called the Photophone

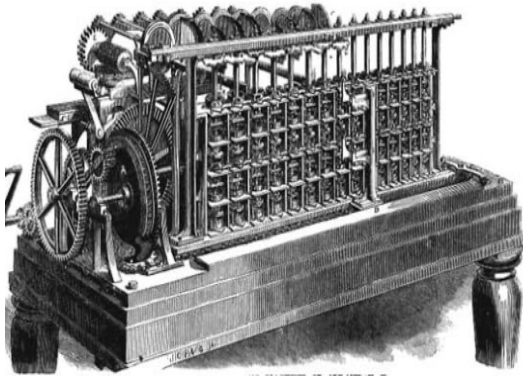
1884 **Comptometer:** The Comptometer which is operated by pressing keys is developed by Dorr E. Felt



Pehr-Georg and Edvard Scheutz's Difference Engine

<https://history-computer.com/pehr-georg-and-edvard-scheutzs-difference-engine/>

History of Computing



Pehr-Georg and Edvard Scheutz's Difference Engine

<https://history-computer.com/pehr-georg-and-edvard-scheutzs-difference-engine/>

Highlights:

- best known for creating the first operational and commercially accessible difference engine.
- an extended, smaller, and better version (£1,638, 1831) of Charles Babbage's Difference Engine (£15,000, 1829).

History of Computing

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William Stanley Jevons

- was a writer, English economist, and logician.
- a major figure, both in Britain and internationally, in political economy and social reform.
- credited with being the first theorist to make economics a mathematical discipline.
- one of the founders of the form of neo-classical economics, which dominates our current economic thinking and political discourse.



The Logic Piano

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History of Computing

1890 **Herman Hollerith:** invents a counting machine which has increment mechanical counters

1895 **Guglielmo Marconi:** develops radio signals

1896 **Tabulating Machine Company:** is formed by Herman Hollerith which later becomes IBM

1898 **Nikola Tesla:** invents the remote control

History of Computing

1906	Lee De Forest: devises the first electronic tube
1911	IBM: IBM is formed on June 15, 1911
1923	Philo Farnsworth: devises the first fully electronic television
1924	John Logie Baird: invents the Electro Mechanical television system Walther Bothe: develops the logic gate
1930	Vannevar Bush: develops a partly electronic Difference Engine (the precursor to the digital computer)

History of Computing

1931	Kurt Godel: publishes a paper on the use of a universal formal language
1937	Alan Turing: develops the concept of a theoretical computing machine
1938	Konrad Zuse: creates the Z1 Computer a binary digital computer using punch tape
1939	George Stibitz: develops the Complex Number Calculator - a foundation for digital computers Hewlett Packard: William Hewlett and David Packard start their computer company John Vincent Atanasoff and Clifford Berry: develop the ABC (Atanasoff-Berry Computer) prototype



History of Computing

1943

Enigma: Adolf Hitler uses the Enigma encryption machine

Colossus: Alan Turing develops the code-breaking machine Colossus

1944

Howard Aiken & Grace Hopper: design the MARK series of computers at Harvard University

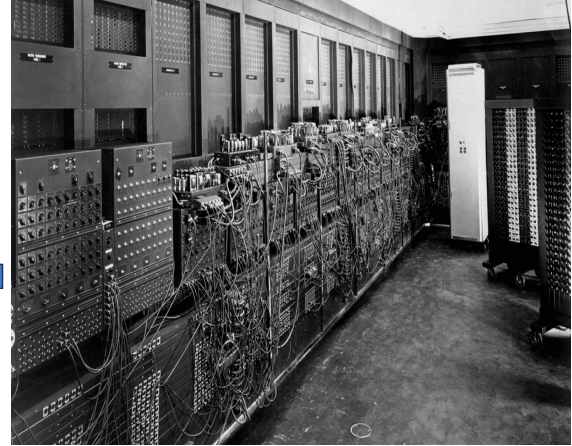
1945

ENIAC: John Presper Eckert & John W. Mauchly develop the ENIAC (Electronic Numerical Integrator and Computer)

Computer Bug: the term computer 'bug' is first used by Grace Hopper

1946

F.C. Williams: develops his cathode-ray tube (CRT) storing device, the forerunner to random-access memory (RAM)



Source: <https://www.simslifecycle.com/>

History of Computing

ENIAC 1 Computer - was developed by John Presper Eckert & John W. Mauchly

Physical Components:

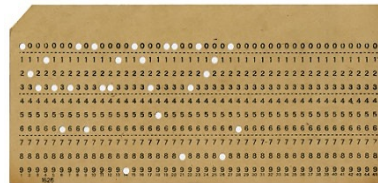
- 17,468 vacuum tubes
- 7,200 diodes
- 1,500 relays
- 70,000 resistors,
- 10,000 capacitors
- CPU size = 2.4m × 0.9m × 30m
- Weight = 27 tons
- Occupies around = 167m²

Input device = IBM card reader

Output device = IBM punch cards

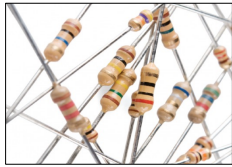


Source: <http://www.gettyimages.com/>



Punch cards

History of Computing



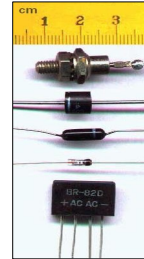
Resistors



Vacuum tubes



Capacitors



Relays

Sources of images:

- <http://electrical-engineering-portal.com/>,
- https://en.wikipedia.org/wiki/Vacuum_tube, learn.sparkfun.com/

History of Computing

1947

Pilot ACE: Donald Watts Davies joins Alan Turing to build the fastest digital computer in England at the time, the Pilot ACE

William Shockley: invents the transistor at Bell Labs

Douglas Engelbart: theorises on interactive computing with keyboard and screen display instead of on punchcards

1948

Andrew Donald Booth: invents magnetic drum memory

Frederic Calland Williams & Tom Kilburn: develop the SSEM "Small Scale Experimental Machine" digital CRT storage which was soon nicknamed the "Baby"

1949

Claude Shannon: builds the first machine that plays chess

Howard Aiken: develops the Harvard-MARK III

History of Computing

1950

Hideo Yamachito: creates the first electronic computer in Japan

Alan Turing: publishes his paper - Computing Machinery and Intelligence which helps create the Turing Test

1951

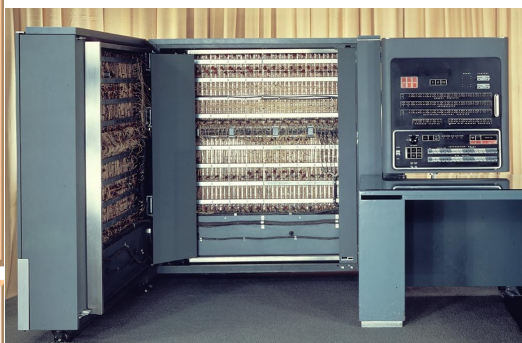
LEO: T. Raymond Thompson and John Simmons develop the first business computer, the Lyons Electronic Office (LEO) at Lyons Co.

UNIVAC: UNIVAC I (UNIVERSAL Automatic Computer I) is introduced - the first commercial computer made in the United States and designed principally by John Presper Eckert & John W. Mauchly

EDVAC: The EDVAC (Electronic Discrete Variable Automatic Computer) begins performing basic tasks. Unlike the ENIAC, it was binary rather than decimal

1953

The IBM 701 becomes available and a total of 19 are sold to the scientific community



IBM 700 Series

History of Computing



UNIVAC I

- 5,200 vacuum tubes,
- 13 metric tons)
- perform about 1,905 op/sec
- running on a 2.25 [MHz](#) clock.

CPU size = 4.3 m x 2.4 x 2.6 m

The complete system occupied more than 35.5 m² (382 ft²) of floor space.

History of Computing

1954	John Backus & IBM: develop the FORTRAN Computer Programming Language
1955	Bell Labs: introduces its first transistor computer
1956	Optical fibre is invented by Basil Hirschowitz, C. Wilbur Peters, and Lawrence E. Curtiss
1957	Sputnik I and Sputnik II: are launched by the Russians
1958	ARPA (Advanced Research Projects Agency) and NASA is formed Silicon chip: the first integrated circuit, or silicon chip, is produced by Jack Kilby & Robert Noyce

History of Computing

1959	Paul Baran: theorises on the "survivability of communication systems under nuclear attack", digital technology and symbiosis between humans and machines
1960	COBOL: the Common Business-Oriented Language (COBOL) programming language is invented
1961	Unimate: General Motors puts the first industrial robot, Unimate, to work in a New Jersey factory
1962	The first computer game: the Spacewar Computer Game is created invented by Steve Russell & MIT
1963	The Computer Mouse: Douglas Engelbart invents and patents the first computer mouse (nicknamed the mouse because the tail came out the end) The American Standard Code for Information Interchange (ASCII) is developed to standardize data exchange among computers

History of Computing

1964

Word processor: IBM introduces the first word processor

BASIC: John Kemeny and Thomas Kurtz develop Beginner's All-purpose Symbolic Instruction Language (BASIC)

1965

Hypertext: Andries van Dam and Ted Nelson coin the term "hypertext"

1967

Floppy Disk: IBM creates the first floppy disk

1969

Seymour Cray: develops the CDC 7600, the first supercomputer

Gary Starkweather: invents the laser printer whilst working with Xerox

ARPANET: The U.S. Department of Defence sets up the Advanced Research Projects Agency Network (ARPANET) with the intention of creating a computer network that can withstand any type of disaster. It becomes the first building block for what the internet is has become today

History of Computing

1970

RAM: Intel introduces the world's first available dynamic RAM (random-access memory) chip and the first microprocessor, the Intel 4004

1971

E-mail: is invented by Ray Tomlinson

Liquid Crystal Display (LCD):is developed by James Fergason

Pocket calculator: is invented by Sharp Corporation

Floppy Disk: is created by David Noble with IBM - Nicknamed the "Floppy" for its flexibility.

1972

First Video Game: Atari releases Pong, the first commercial video game

The CD: the compact disc is invented in the United States.

1973

Robert Metcalfe and David Boggs: creates the Ethernet, a local-area network (LAN) protocol

Personal computer: The minicomputer Xerox Alto (1973) was a landmark step in the development of personal computers

Gateways: Vint Cerf and Bob Kahn develop gateway routing computers to negotiate between the various national networks

History of Computing

1974

SQL: IBM develops SEQUEL (Structured English Query Language) now known as SQL

WYSIWYG: Charles Simonyi coins the term WYSIWYG (What You See Is What You Get) to describe the ability of being able to display a file or document exactly how it is going to be printed or viewed

1975

Portable computers: The Altair 8800 is developed by Micro Instrumentation and Telemetry Systems (MITS) and the first personal home computer is released on 23 Jan, 1975

Microsoft Corporation: is founded April 4, 1975 by Bill Gates and Paul Allen to develop and sell BASIC interpreters for the Altair 8800

1976

Apple: Apple Computers is founded by Steve Wozniak and Steve Jobs on 1 Apr 1976 (April Fools Day)

1977

Apple Computer's **Apple II**, the first personal computer with colour graphics, is demonstrated

MODEM: Ward Christensen writes the programme "MODEM" allowing two microcomputers to exchange files with each other over a phone line



History of Computing

1978

Magnetic tape: The first magnetic tape is developed in the US

1979

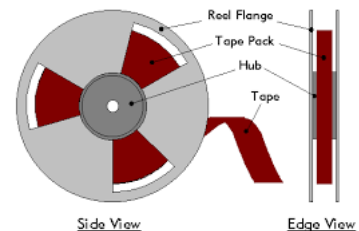
Over half a million computers are in use in the United States

1980

Paul Allen and Bill Gates: IBM hires Paul Allen and Bill Gates to create an operating system for a new PC. They buy the rights to a simple operating system manufactured by Seattle Computer Products and use it as a template to develop DOS

1981

Microsoft: MS-DOS Computer Operating System increases its success



History of Computing

1982

WordPerfect: the WordPerfect Corporation introduces WordPerfect 1.0 a word processing program

Commodore 64: becomes the best-selling computer of all time.

SMTP: SMTP (Simple Mail Transfer Protocol) is introduced for the first time

1983

More than 10 million computers are in use in the United States

Domain Name System (DNS): is pioneered by Jon Postel, Paul Mockapetris and Craig Partridge. Seven 'top-level' domain names are initially introduced: edu, com, gov, mil, net, org and int.

Microsoft Windows: is introduced eliminating the need for a user to have to type each command, like MS-DOS, by using a mouse to navigate through drop-down menus, tabs and icons

1984

Apple Macintosh: Apple introduces the Macintosh with mouse and window interface on 24 Jan, 1984

Cyberspace: William Gibson coins the word cyberspace when he publishes Neuromancer

Dell Computer is founded in Austin Texas on 3 May, 1984



History of Computing

1985

Paul Brainard introduces Pagemaker for the Macintosh creating the desktop publishing field

Nintendo: The Nintendo Entertainment System makes its debut.

Microsoft Windows 1.0 is introduced on 19 Nov, 1985 and is initially sold for \$100.00

1986

More than 30 million computers are in use in the United States.

1987

Microsoft introduces **Microsoft Works**

Perl: Larry Wall introduces Perl 1.0

1988

Over 45 million PCs are in use in the United States.

History of Computing

2001

Xbox: Bill Gates introduces the Xbox on 7 Jan, 2001

Windows XP: is launched on 20 Sep, 2001

2002

Approximately 1 billion PCs have been sold

PayPal is acquired by **eBay**

2003

iTunes is launched on 9 Jan, 2003

Skype is launched on 20 Aug, 2003

2004

Facebook is first launched on 20 Sep, 2004

Firefox 1.0 is released on 9 Nov, 2004

Blizzard's **World of Warcraft** game, the most popular and successful [MMORPG](#) is released 23 Nov, 2004

History of Computing

2005

The [blu-ray](#) is first announced and introduced on 4 Jan, 2006.

YouTube begins on 14 Feb, 2005

Xbox360 is released on 20 Feb, 2005

eBay acquires **Skpe** on September 12, 2005

2006

iPhone: the first iPhone is released on 29 Jun, 2006

SCE releases **Play Station 3** on 11 Nov, 2006

Nintendo releases **Wii on** 19 Nov, 2006

2007

Windows Vista is released 30 Jan, 2007

Netbook: Asus released first Eee PC (netbook) on 16 Oct, 2007

2008

First **Android OS and Android Smartphone** is released on 21 Oct, 2008

History of Computing

2009

Windows 7 is released on 22 Oct, 2009

2010

The iPad is released on 3 Apr, 2010

2011

Microsoft exceeds \$1 billion in sales and becomes the first company to do so

2012

Apple introduces the **iPhone 5**

Microsoft Windows 8 and **Microsoft Surface** is released October 26, 2012

Apple introduces the **iPad mini** October 23, 2012

The number of **smart phones** worldwide reaches 1 billion

History of Computing

2013

Apple introduces **iOS7** on September 18, 2013

Microsoft releases **Windows 8.1** on October 18, 2013

Sony releases the **PS4** November 15, 2013

Apple outsells Windows PCs globally for the first time

2014

The Nokia deal with Microsoft is completed April 25 making Nokia now **Microsoft Mobile** in a deal totalling \$7.17 billion

Apple introduces the new iPhone 6, iPhone 6 Plus, and Apple Watch on September 9

Microsoft announces on September 30 the next version of Windows, which will be called **Windows 10**

Android version 5.0 (Lollipop) released November 3

2015

Apple officially releases the **Apple Watch** on April 24

Microsoft releases **Windows 10** on July 29

Android version 6.0 (Marshmallow) released October 5

Microsoft releases the **Surface Book** on October 26

History of Computing

2016

Pokémon GO is released July 6, for Android phones and iPhone
 Google releases **Android version 7.0 (Nougat)** August 22
 On September 16, Apple announces the new **iPhone 7 and iPhone 7 Plus**

2017

Microsoft introduced the **Surface Pro 2017** (fifth gen) and the **Surface Laptop** on June 15
 Apple introduced the **iPhone 8, iPhone 8 plus**, with a new edge-to-edge iPhone X with Face ID on September 12
 On November 14 Mozilla announced the release of **Firefox Quantum**
 The Defense Advanced Research Projects Agency (DARPA) begins developing a new "**Molecular Informatics**" program that uses molecules as computers.

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2018

Google introduces the **Google Pay** service for all Android and iOS devices on January 8
 Apple introduces the **Apple HomePod**
 The European Union GDPR (**General Data Protection Regulation**) law comes into effect on May 25
 On June 4, Microsoft announces it would **acquire GitHub** for \$7.5 billion
 On October 28 BM announce its plans to **acquire Red Hat** for approximately \$34 billion

History of Computing

2019

Computed axial lithography, a **3-D printing technology** is developed by UC Berkeley and Lawrence Livermore National Laboratory is announced in January

On January 9, Lexar announces the first SD card which can **store 1 terabyte**

The first **folding smartphones** are introduced in February by Samsung (Galaxy Fold) and Huawei (Mate X)

Disney+ is introduced

IBM Q System One, the **first commercial quantum computer** from IBM, is announced

Google makes cloud gaming mainstream with **Stadia**

IPv4 internet addresses are exhausted

Google begins drone based deliveries via **Google Wing**

History of Computing

2020

On January 6, Lenovo announced the **Yoga 5G**, a 5G-capable laptop

On January 14, **Windows 7 reached EOL** (end-of-life) and Microsoft ceases providing any support or security updates for Windows 7

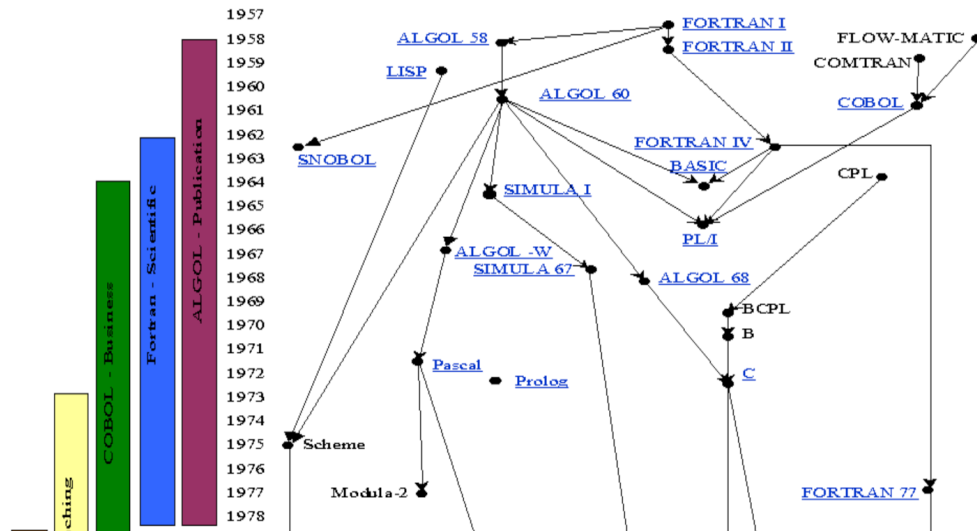
On January 16, Alphabet, the parent company of **Google**, **joined the "four-comma club,"** achieving a market valuation over \$1 trillion. The other two members of the club at the time were Apple (\$1.3 trillion) and Microsoft (\$1.2 trillion). The next closest company, Amazon (\$930 billion), briefly joined the club in 2018

In early February, the outbreak of the COVID-19 coronavirus causes significant **disruption to the worldwide technology industry**. Google announces the temporary closure of all offices in mainland China, and all Apple stores in China are also closed indefinitely. Companies including Qualcomm, Nintendo, and Facebook Oculus announce major losses in production from their factories located in China. The Chinese government reports a 50% decrease in smartphone sales

On February 4, Twitter announces a commitment to **detect and label deep-fakes**, and other deceptive media, on its social media platform

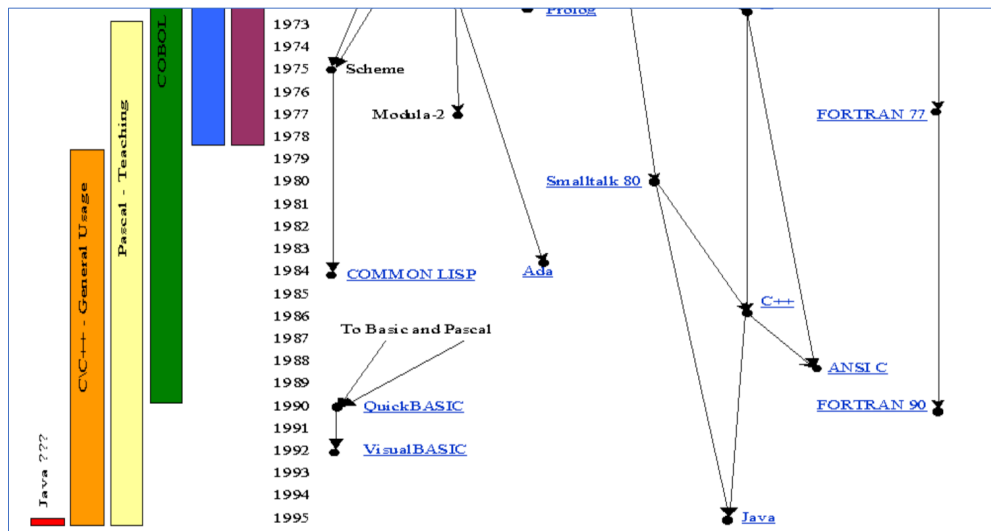
In May, **Giphy** is **purchased by Facebook**

History of Programming Languages



<https://web.archive.org/web/20070203095546/http://www.cs.iastate.edu:80/~leavens/ComS541Fall97/hw-pages/history/>

History of Programming Languages



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